
IAL Biology Unit 6

Rapid Revision

WBI16 — Practical Skills in Biology II

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* study note
(19 past papers and mark schemes, October 2019 – June 2026)

50 marks · 1 h 20 min · 4 questions: two method (~10 each), one data analysis (~14), one planning (15–18)

A* rule: grade A overall ($\geq 480/600$ UMS) **and** $\geq 90\%$ across Units 4, 5 and 6 combined ($\geq 270/300$). The A* is decided entirely by A2 performance.

TRIAGE — WHAT THIS PAPER REWARDS

WHERE THE MARKS ARE

Planning question · "describe an experiment" · null hypothesis, table and graph · a statistical test · variables with control and effect · risk — **every paper**.

WHAT THE MARK SCHEMES REWARD THAT STUDENTS MISS

1. Quantities, not "amount" — volume, mass, number, with a unit. **2.** A **stated number**: "5, 15, 25, 35 and 45 °C", not "different temperatures". **3.** **How** a variable is controlled, not just what. **4.** Effects must be **directional**. **5.** Graph and test must match the design. **You have 1.6 min per mark — more than Units 4 and 5. Use it to write every method point.**

UNIT 6 — SHEET 15 · THE PLANNING QUESTION (15–22 marks, every paper)

Part	Marks	What scores
(a)	2–3	Preliminary work. Always a " find a suitable... " statement — never a method. "Ignore practise the method"; "Do not award method statements". Say what the "suitable" is for: "a suitable temperature for measurable respiration to occur ".
(b)	8–10	The method. Marked as "an answer that includes eight of the following" — every separate correct point is a mark .
(c)	3–4	Table + graph + named test (see sheet 17).
(d)	2–4	Limitations — one of each of the three kinds below.

THE METHOD — THE UNIVERSAL MARK POINTS. WRITE ONE SENTENCE EACH.

1. DV stated clearly with how it is measured — "volume of O₂ released per minute". "Ignore amount"; "Ignore rate" alone; "**do not piece together the dependent variable**".
2. IV with a range of stated values — "0, 0.5, 1.0, 1.5 and 2.0 mol dm⁻³". "Different temperatures" scores nothing.
3. How the IV is changed. **4.** Apparatus for measuring the DV (respirometer, photosynthometer, colorimeter — labelled diagram accepted).
5. Standardise the organisms — same mass / length / strain / source.
6. Acclimatise / equilibrate before measuring. **7. A stated time** for the measurement.
8. Calculate a rate — volume ÷ time, 1 ÷ time, $\pi r^2 d \div t$.
9–11. Two controlled variables AND how each is controlled.
12. A control for comparison. **13.** Aseptic technique if microbes. **14. Repeats → mean AND standard deviation.** **15.** Welfare if animals.

Limitations — give one of each kind: (i) measuring the DV is hard (small gas volumes; O₂ dissolves in water; judging a colour end point; zones not circular) · (ii) a variable is hard to control (germinating seeds and lamps produce heat; age/viability/genetic variation) · (iii) the biology changes during the experiment (aerobic → anaerobic; waste builds up; gases expand when warm; CO₂ is soluble). "Ignore bubbles / escape of gas unqualified" — say **what** is hard and **why it matters**.

UNIT 6 — SHEET 16 · VARIABLES, CONTROL METHODS & RISK

THE THREE-PART VARIABLE QUESTION — 3 TO 5 MARKS EVERY PAPER

1. Name the variable in the **right category**. A biotic answer on an abiotic line scores 0 and **negates the line**. **Write one variable per line** — only the first answer on each line is marked.
2. Say **HOW** it is controlled — the method, not just the variable.
3. The effect if uncontrolled **must be DIRECTIONAL**. "It would affect the results" = 0. "A higher temperature would increase enzyme activity, **so more CO₂ would be produced**" = 1.
The safe fallback that is always on the list: "**the results would not be valid**". Never write "not reliable / not accurate" — those are ignored.

Variable	Control method the schemes accept
Temperature (in solution)	Thermostatic water bath or incubator — " ignore waterbath unqualified "
Temperature (room / animal)	Air-conditioned or thermostatically controlled room
Temperature (aquatic plant)	Thermostatic water bath; heat shield between lamp and plant — "ignore AC room"
pH	Buffer at a stated pH (in the field: select sites of similar pH, not a buffer)
Light intensity / wavelength	Same wattage bulb at a fixed measured distance ; light meter / same filter
Mass / length of material	Balance; ruler; cork borer — "IGNORE size"
Age of organism	Hatched at the same time; seeds from the same batch
Stimulus (touch)	Same glass rod / cotton bud, same force, same person

Things the schemes ignore as variables: "nutrients" (say minerals / nitrate / glucose) · "size" · "type" · "genetic variation" · "stress".

Risk = 2 marks: a named risk + a matching reduction. Microbiology → infection by pathogens ("not contamination unqualified") → aseptic technique, tape the lid, incubate below 30 °C. Spirometer → infection from the mouthpiece → sterilise or replace it. Fibre tensile test → snapping fibre / falling masses → goggles and a sand tray. "**Wear a lab coat**" and "**be careful**" **score nothing** — name the item and match it to the risk.

UNIT 6 — SHEET 17 · TABLE, GRAPH & THE FOUR TESTS

THE PAIRING RULE — A MISMATCH COSTS THE MARK

Two groups compared (two varieties, with vs without, trained vs untrained) → **bar graph + t-test** (test for **difference**).
Several values of a continuous IV (5 light intensities, 5 temperatures, ages) → **line or scatter graph + Spearman's rank** (test for **correlation**).
Counts in categories (choice chamber, fruiting / not fruiting) → **bar chart + chi-squared**.

The table — 3 marks: suitable format with raw data · **headings with units, unit in the heading only** ("Do not accept units in the body of the table") · **means to the same number of decimal places as the data**. An extra column **negates MP1** — no "trial number" column. No calculated rate in a raw-data table.

Test	Formula	Decision
Student's t 12 of 19 papers — the default	$t = \frac{ \bar{x}_1 - \bar{x}_2 }{\sqrt{(s_1^2/n_1 + s_2^2/n_2)}}$ df = $n_1 + n_2 - 2$	$t >$ critical → reject H ₀ (significant difference). The paper gives you s² — do not square it again . "Ignore minus sign."
Spearman's rank 5 papers	$r_s = 1 - \frac{(6\sum d^2)}{(n(n^2 - 1))}$ Rank each variable separately; ties share the mean rank	$r_s >$ critical → reject H ₀ (significant correlation). State positive or negative . At $n = 5$ the critical value is 1.000 — only a perfect correlation is significant.
Chi-squared 3 papers	$\chi^2 = \sum (O - E)^2 \div E$ df = categories – 1; E = total ÷ number of categories	$\chi^2 >$ critical → reject H ₀ . Used because the data are categorical / frequency counts .

Mann-Whitney U 2 papers	$U = R - n(n + 1)/2$ for each group; R = sum of ranks when both groups are ranked together	OPPOSITE WAY ROUND: significant if the smallest U is $\leq$ the critical value . Compares medians .
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THE DECISION SENTENCE — NEVER OMIT THE COMPARISON

"The calculated value of **X** is **greater / less than** the critical value of **Y** at $p = 0.05$, **so we reject / accept the null hypothesis**: there is (no) significant difference / correlation between ___ and ___."

"Must state the calculated value is more / less than the critical value" — a decision without the comparison scores **nothing**.

Null hypothesis = there is **no significant difference / correlation** between the two named things.

Error bars / SDs: overlap $\rightarrow$ not significant; no overlap $\rightarrow$ significant. Always name which groups — "don't accept overlapping SDs are not significant unqualified". Add "a statistical test would be needed to confirm".

Other calculations: % change = $(\text{new} - \text{original}) \div \text{original} \times 100$ · magnification = image $\div$ actual (convert units first) · rate = change $\div$ time **with units** ("Not dm^3/s " — write $\text{dm}^3 \text{ s}^{-1}$).

THE REJECTIONS BANK — THE MARKS MOST OFTEN LOST

Do not write	Write instead	
	"Amount"; "different temperatures"	Volume / mass / number with a unit; stated values
	"It would affect the results"	A directional effect on the DV
	"Water bath" unqualified; "be careful"; "wear a lab coat"	Thermostatic water bath; a specific reduction matched to the risk
	Units in the body of the table; an extra column	Units in the heading only ; no trial-number column
	A decision with no comparison to the critical value	"X is greater than the critical value of Y, so reject H_0 "

WALKING INTO THE EXAM

Unit 6 — you have 1.6 min per mark, so write every method point; state numbers; say **how** each variable is controlled and give a **directional** effect; match graph to test; never give a decision without comparing to the critical value.